

Cambridge Lower Secondary Checkpoint

CANDIDATE
NAME

CENTRE
NUMBER

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CANDIDATE
NUMBER

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SCIENCE

0893/01

Paper 1

October 2025

45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should show all your working in the booklet.
- You may use a calculator.

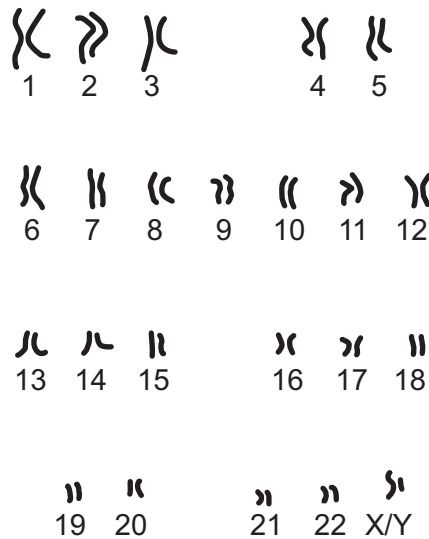
INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

1 Blessy is learning about DNA.

She finds this diagram of strands of DNA that have been taken from the nucleus of a human cell.



(a) Write down the name of these strands of DNA.

..... [1]

(b) Small sections of these strands determine characteristics such as eye colour.

What are these small sections of DNA called?

..... [1]

(c) Blessy makes this conclusion about the strands of DNA,

‘These strands of DNA were taken from a man and not a woman.’

Tick (✓) to show if Blessy is correct.

Yes ☐ No ☐

Explain your answer.

.....
 [1]

2 Look at the electronic structures of some atoms, **A** to **F**.

2.8.1

A

2.7

B

1

C

2.8.8.1

D

2.3

E

2

F

(a) Which electronic structure shows an atom of an element in **Group 3** of the Periodic Table?

.....

[1]

(b) Which electronic structure shows an atom of an element in **Period 4** of the Periodic Table?

.....

[1]

(c) Which electronic structure shows an atom of an element in **Group 8** of the Periodic Table?

.....

[1]

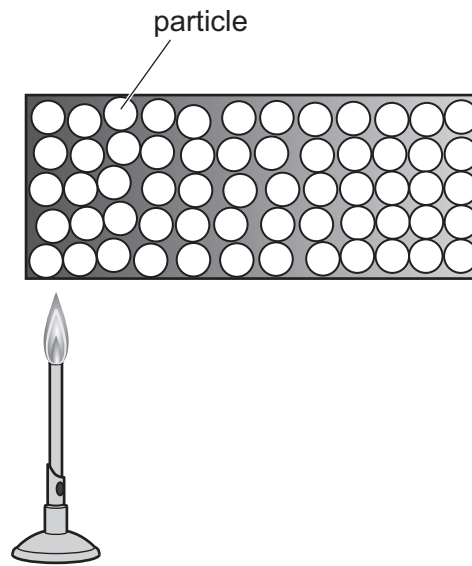
(d) Which **two** electronic structures show atoms whose elements are in the **same group** of the Periodic Table?

..... and

[1]

- 3 This question is about the transfer of thermal energy.

The diagram shows a Bunsen burner heating a solid object at one end.



Conduction transfers thermal energy through a solid object.

Complete these sentences about conduction.

The Bunsen burner heats one end of the solid object.

The particles in the object gain energy and more.

The particles with nearby particles.

This passes the energy from the end of the object to the end of the object.

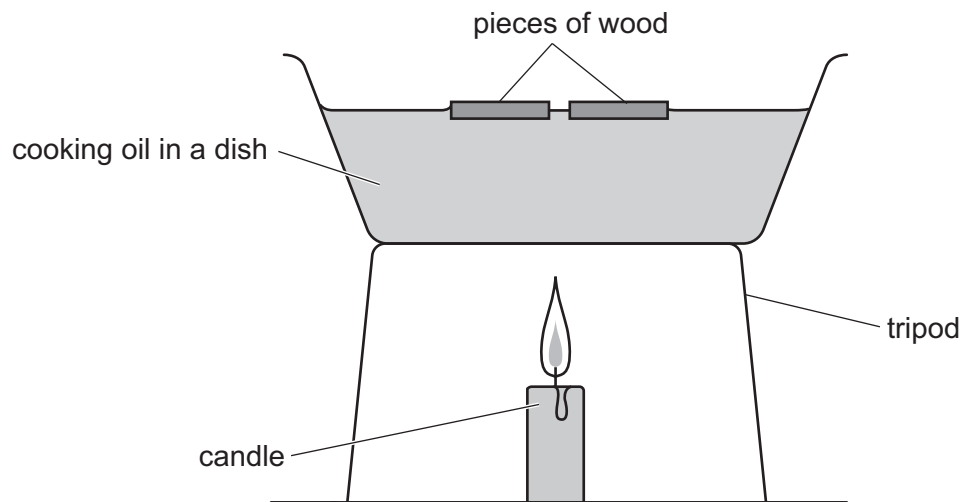
[3]

- 4 Mia uses a model to show how tectonic plates move.

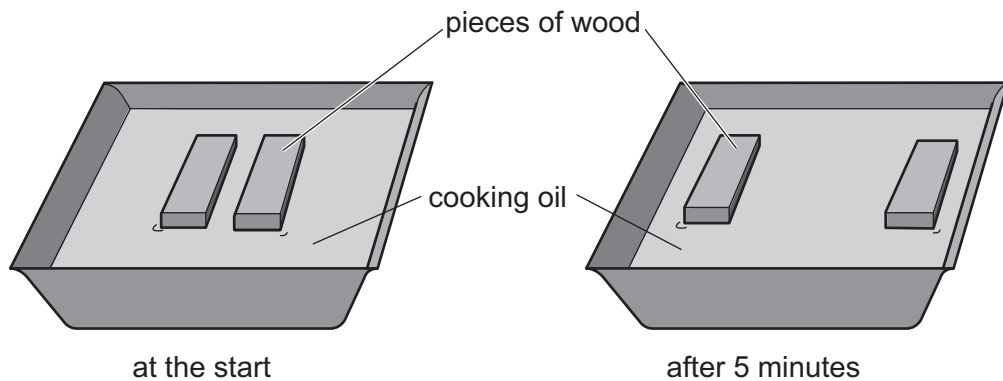
Mia:

- floats two pieces of wood on cooking oil in a dish
- places a candle below the dish
- lights the candle.

Look at the assembled equipment.



Mia looks down at the assembled equipment and makes these drawings.



Explain why the two pieces of wood move apart.

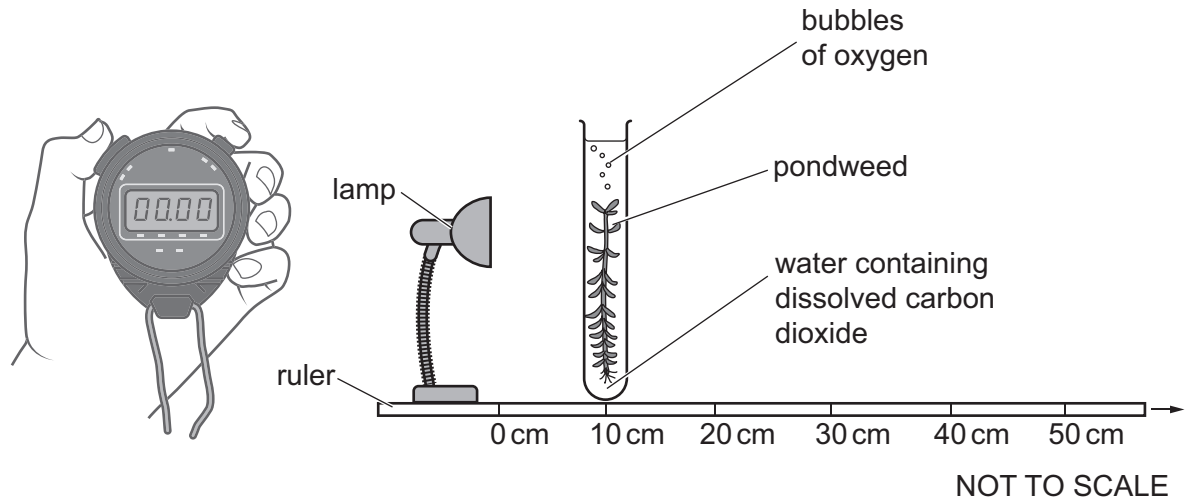
.....

.....

..... [2]

- 5 Jamila investigates the effect of light intensity on the rate of photosynthesis.

The diagram shows the assembled equipment she uses.



- (a) Describe how Jamila uses this equipment to investigate how light intensity affects the rate of photosynthesis.

.....

.....

.....

.....

..... [3]

- (b) State **one** variable Jamila controls in her investigation.

..... [1]

- (c) Predict the effect of changing the light intensity on the rate of photosynthesis.

.....

..... [1]

- (d) Jamila does each experiment of her investigation three times.

She then calculates the mean rate of photosynthesis.

Explain why this improves her investigation.

.....

.....

..... [2]

- 6 Ahmed prepares some copper sulfate crystals.

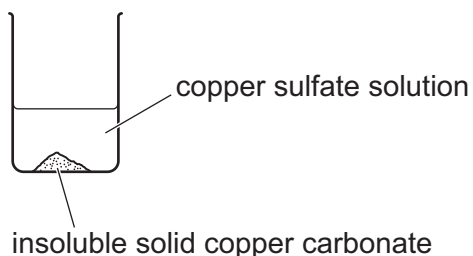
He reacts solid copper carbonate with dilute sulfuric acid in a beaker.

Copper sulfate solution, carbon dioxide and water are made.

- (a) Write the **word** equation for this reaction.

..... [2]

- (b) At the end of the reaction, copper sulfate solution and some insoluble solid copper carbonate are left in the beaker.



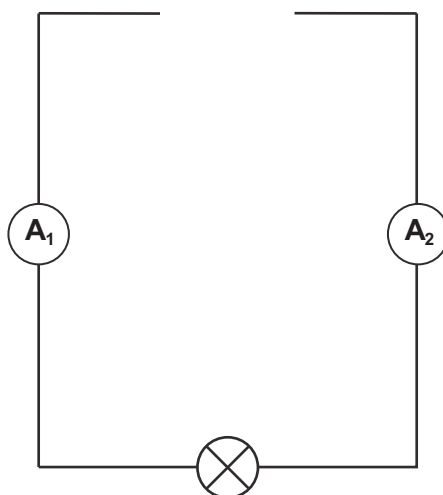
Describe how Ahmed separates the copper sulfate solution from the insoluble solid copper carbonate.

..... [1]

- (c) Describe how Ahmed makes copper sulfate crystals from the copper sulfate solution.

.....
.....
.....
..... [2]

7 Yuri makes an electrical circuit.



(a) Yuri puts **one** cell in the gap in his circuit.

Draw the standard conventional symbol for a cell in the gap in his circuit.

[1]

(b) The reading on ammeter **A₁** is 0.55 A.

State the reading on ammeter **A₂**.

..... A

[1]

(c) Yuri measures the voltage across the lamp.

Draw the position of a voltmeter on his electrical circuit to measure the voltage across the lamp.

Use the correct standard conventional symbol for a voltmeter.

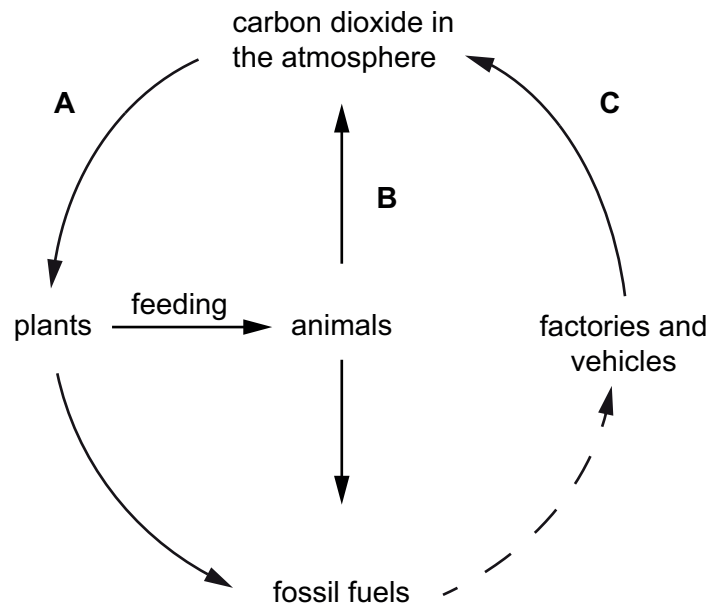
[2]

(d) Yuri adds another lamp in series with the first lamp.

What happens to the reading on ammeter **A₁**?

..... [1]

8 The diagram shows a simplified carbon cycle.



(a) Write down the names of the processes, **A**, **B** and **C**.

process **A**

process **B**

process **C**

[3]

(b) Scientists are concerned about **climate change**.

Write down **one** possible impact of climate change.

..... [1]

- 9 The picture shows two butterflies.

The butterflies are the same species.



- (a) (i) Write down **two** ways the butterflies show variation within the species.

Use information seen in the pictures.

1

2

[2]

- (ii) The environment causes variation.

Which **other** factor causes variation within a species?

..... [1]

- (b) Male butterflies fertilise the eggs from female butterflies.

Describe what happens during fertilisation.

.....

.....

..... [2]

10 This question is about density.

- (a) Solid **A** has a mass of 20 g and a volume of 4 cm³.

Calculate the density of solid **A**.

Include the unit of density.

density of solid **A** = unit [3]

- (b) Another solid, **B**, is added to water.

Solid **B** does **not** react with water or dissolve in water.

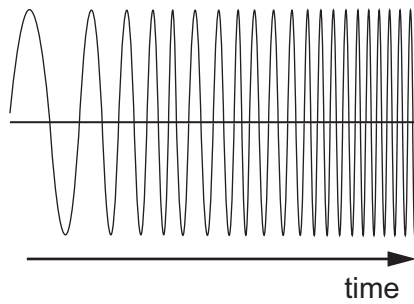
Solid **B** has a density of 0.8 g/cm³.

What happens when solid **B** is put into water?

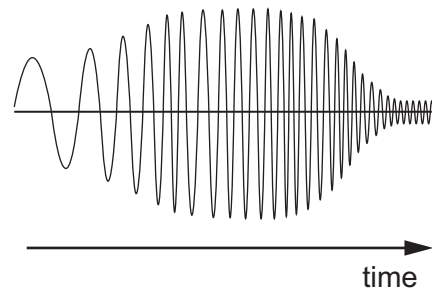
.....
Explain why.

.....
..... [2]

11 Oliver looks at two different sound waveforms.



sound waveform **A**



sound waveform **B**

(a) Look at sound waveform **A**.

Which feature remains the same and which feature changes in waveform **A**?

remains the same

changes

[2]

(b) Look at sound waveform **B**.

Oliver listens to the sound produced by waveform **B**.

Describe what he hears.

.....

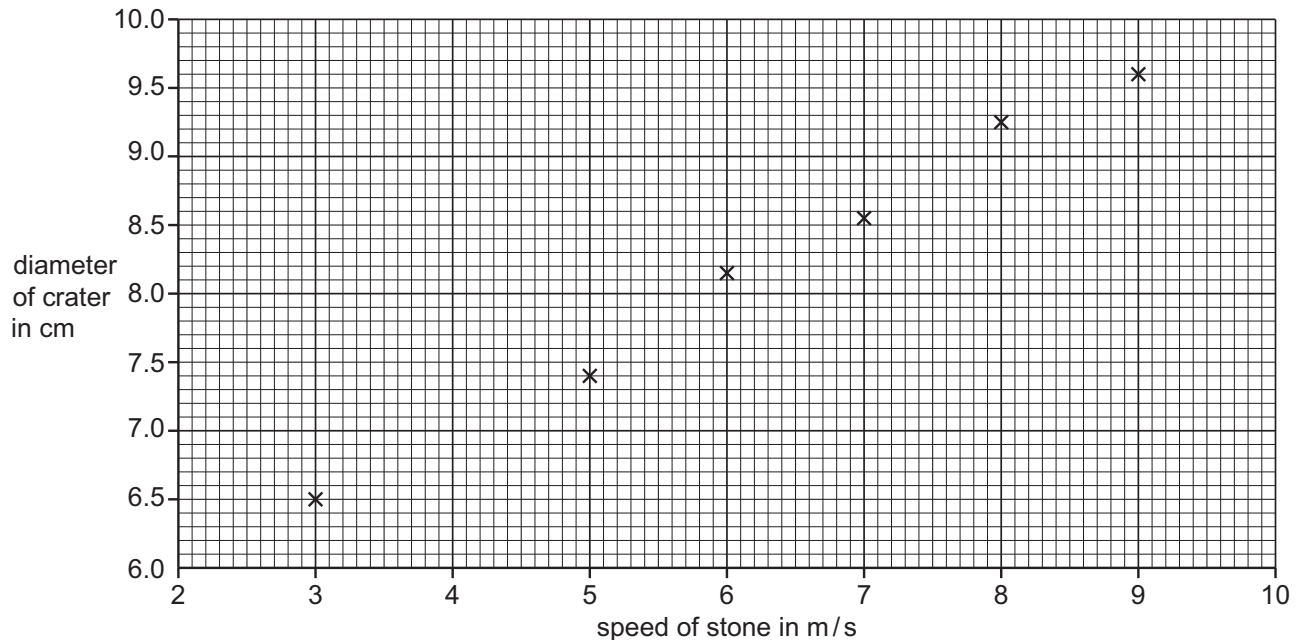
 [2]

12 When an asteroid hits the Earth's surface the asteroid forms a crater.

Priya investigates how the speed of an asteroid affects the diameter of the crater formed.

She drops a stone into sand and measures the diameter of the crater.

Her results are plotted on the graph.



(a) A stone has a speed of 4m/s.

The stone forms a crater with a diameter of 6.9 cm.

Plot this result on the graph.

[1]

(b) Draw a line of best fit for the data on the graph.

[1]

(c) Complete the sentence about the relationship between the speed of the stone and the diameter of the crater.

As the speed of the stone, the diameter of the crater

[1]

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The Periodic Table of Elements

Group																	
1	2	1 H hydrogen 1						3	4	5	6	7	8				
Key atomic number atomic symbol name relative atomic mass																	
3 Li lithium 7	4 Be beryllium 9																
11 Na sodium 23	12 Mg magnesium 24																
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —
lanthanoids																	
actinoids																	